Paper opportunities in index calculus and fall degrees: which paper to submit

2026-10-29

Paper opportunities in index calculus and fall degrees

What this is. A search across `crypto`, `crypto-autoresearcher` and `cryptanalysis` for the best paper the programme could submit. Scope, as the owner set it: index calculus and first/last fall degrees (ffd, lfd). The ECC2K-130 GPU paper is out of scope because it is already public as IACR ePrint 2026/2469 (14 October 2026).

This is a read-only assessment, plus one re-verification (§6). It ran no experiment, changes no research state, and promotes nothing. Candidate results are quoted at the tier their sources give.

1. Recommendation

Submit one paper on the binary Semaev descent: "The first fall degree is constant, the last fall degree is not: proofs and checkable certificates for summation-polynomial systems." Post it to IACR ePrint, then submit to the *Journal of Mathematical Cryptology* (where Kousidis–Wiemers 2019 appeared) or to *Designs, Codes and Cryptography.*

Its core already exists in `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/` (721-line theorem dossier, two independent implementations, a certificate checker). Fold in the explicit-membership "guarded" chain theorems from `crypto:research/guarded_fall_degree_20261028/`.

Why this one:

1. **It answers a stated open question.** Huang–Kosters–Yeo (CRYPTO 2015) and Kousidis–Wiemers (JMC 2019) both name rigorous bounds on the last fall degree of summation-polynomial Weil descent as open. The only post-2020 lower bounds I found are for other systems (iterated MiMC-type systems, arXiv 2310.03637). Huang 2103.07282 and 2311.02804 give upper bounds only.
2. **It replaces solver observations with intrinsic, independently checkable statements.** Kosters–Yeo (2015) and Kousidis–Wiemers (2019) already saw Magma/F4 *step degrees* reach 5. Those are properties of a run. The dossier's statements are properties of the system: $1 \notin W_4(F)$, hence $d_{\mathrm{LFD}} \geq 5$ and *every* F4 run reaches degree ≥ 5. A short checker that shares no code with the producer verifies each one.
3. **It has proofs, not only data.** All proved for every n and V:
   - an exact reformulation of S_3;
   - $d_{\mathrm{ff}} = 2$ for the whole chained system and every t, where Semaev's ePrint 2015/310 states 4;
   - $d_{\mathrm{LFD}} \leq k+2$, and $(t-1)k+t$ for chains;
   - $d_{\mathrm{LFD}} \geq 3$ under checkable conditions.

   The guarded study adds $(d_{\mathrm{ff}}, d_{\mathrm{LFD}}) = (4,5)$ *exactly* for every $p > 3$ at $B = 3$, the binary analogue (exact except the even-degree cube orbit), and $\max(5, B+1) \leq d_{\mathrm{LFD}} \leq 3B-4$.
4. **It is honest about scope.** Its §5 already says it proves no growth to infinity and makes no cost claim.

Figure 1 shows the central separation: the first fall degree stays flat while the last fall degree climbs.

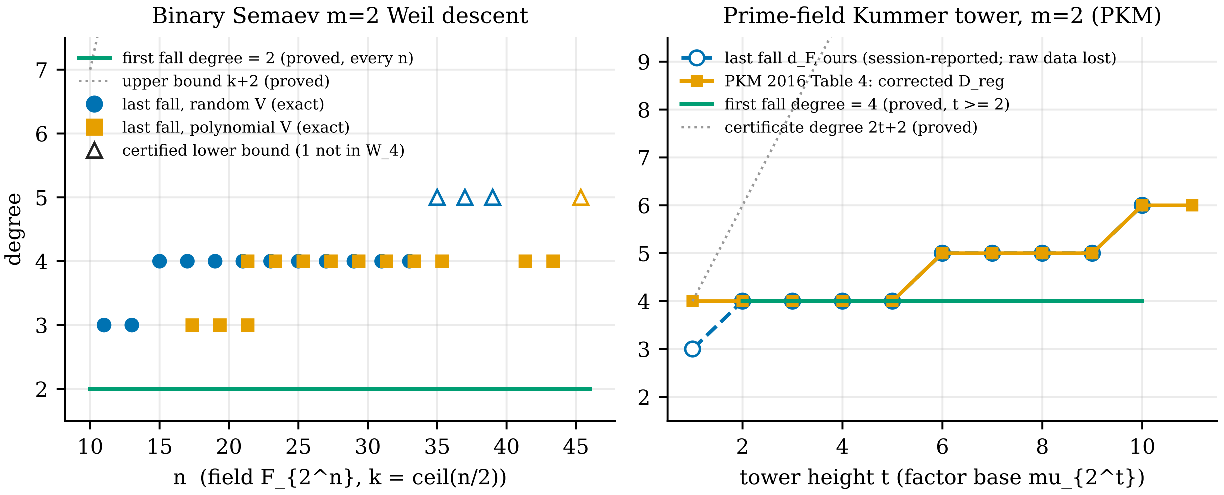

Figure 1: Figure 1. Left: binary Semaev $m = 2$ descent. First fall degree 2 (proved), exact last fall degrees per instance, and certified lower bounds $d_{\mathrm{LFD}} \geq 5$ (open triangles) at $n = 35, 37, 39$ (random V) and $n = 45$ (Semaev's default polynomial V). Right: prime-field Kummer tower ($m = 2$). The session-reported d_F series lies on PKM 2016 Table 4's corrected degree of regularity. The growth is prior art; the proved constant ffd $= 4$ and the $2t + 2$ certificate are ours. Values and sources: `figures/data.json`.

2. What must be true before submission (gating)

#	item	why	effort
G1	**Publish the certificates.** Only their SHA-256 hashes are committed (`results/certificates.txt`); the files are not in any repository. Regenerate them (§6 shows this works) and deposit them in an archive (P45a is about 0.3 GB).	A referee must be able to run `cert_check.py`.	1–2 days of compute
G2	**More instances per n.** Only 2 instances per n at $n \geq 33$; 1 polynomial-V instance at $n \geq 41$. Certify about 10 per n for $n = 33 \ldots 45$, and push to $n = 47 \ldots 51$ to see whether 5 becomes 6.	Makes the step $4 \to 5$ a population statement rather than a handful of instances.	1–3 weeks on a large-memory host (P45a closure took about 6,000 s)
G3	**Expert proof review** of Thm 2.1, Cor 2.2/2.7, Prop 2.6, Thm 2.5, Prop 3.7, and the guarded B1–B4 proofs.	All are written proofs; no Lean beyond small identities.	external

#	item	why	effort
G4	**One fall-degree convention.** Use Caminata–Gorla closure $V_{F,d}$ for d_{LFD} and CG Def. 1.3 for d_{ff}. State how the Petit–Quisquater reading differs; the 2026-10-22 replay counterexample shows the reading matters.	Reviewers in this area check definitions first.	days
G5	**Literature section, read in full.** Read: Semaev 2015/310 (latest version), Kosters–Yeo 1503.08001, HKY 2015/573, HKYY 1505.02532, Kousidis–Wiemers (JMC 2019), Huang–Petit–Shinohara–Takagi 2015/358, Caminata–Gorla, Gorla–Müller–Petit, Huang 2103.07282 and 2311.02804. State plainly that "Assumption 1 fails" was already observed empirically (KY 2015 at $n = 45$; KW 2019). The new part is the intrinsic certificate and the proofs.	Avoids a novelty overclaim.	1 week
G6	**Cite B5/B6 correctly or drop them.** The Boolean upper bounds $3\ell + 1$ and $3\ell + 2$ are weaker than the dossier's own $2k + 3$ and are standard brute-force PC refutations.	Keep the paper's claims tight.	hours

Stretch, which would move the venue up. A *proved* lower bound on d_{LFD} that grows with n (or k) for a summation-polynomial system would resolve the open question outright, and would merit CRYPTO or EUROCRYPT. Semaev's own abstract concedes his asymptotics survive if the regularity degree is $o(\sqrt{n \ln n})$. So only a growing lower bound, not the certificates, bears on the subexponential claim. The dossier's §4 heuristic and the regular "$4n$ new cubics per round, then stall" closure pattern are the starting points. Treat this as research, not as a submission requirement.

3. Second paper: prime-field Kummer towers (PKM)

`crypto:research/prime_fall_degree_20261027/`, written proofs D1–D8:

- **ffd** $= 4$ **for every tower height** $t \geq 2$ (D3, via $\det M = -2g^6(3x_R^2 + a) \neq 0$);
- certificate degree $2 + \operatorname{popdeg}(f^{-1}) \leq 2t + 2$ (D4);
- **presentation dependence**: the naive univariate encoding has $d_F \geq 2^t$ (D5);
- a single summand has $d_F = 2$ for every t (D8).

Two facts limit it today:

1. **The measured growth is prior art.** PKM 2016's own Table 4 (prime case, $p - 1$ subgroups) reports corrected D_{reg} 4,4,4,4,4,5,5,5,5,6,6 for $n_1 = 1..11$. Our session-reported d_F for $t = 1..10$ is 3,4,4,4,4,5,5,5,5,6 (Figure 1, right).
2. **The raw data is lost.** `runs/`, `TABLE.md`, `RESULTS.md` and four `derived/*.md` files were never committed (`RESTART.md`). Stage A at $t \geq 11$, stage C and the registered prediction $d_F = 4 + \lfloor (t-2)/4 \rfloor$ are untested.

Best use: regenerate stage A for $t = 1..14$ with closure certificates (as in G1), score the registered prediction, and publish either as a short JMC/DCC paper or as one section of the main paper. Its contribution is the *proved* constant ffd beside a growing d_F in the setting PKM introduced, plus the presentation-dependence theorem.

4. Short notes worth an ePrint each

- **Nagao Lemma 4** (`crypto-autoresearcher:papers/nagao-lemma4-note/`).
  - Under the equality reading, $d_F \leq d'_F$ fails: $\{X^2Y, XY + X\}$ gives $d'_F = 2 < d_F = 3$.
  - 94 of 400 random multilinear pairs violate it.
  - Owed: typeset it, have a human check it, and contact the author.
- **Fall-degree definitions matter** (`crypto-autoresearcher:docs/reports/2026-10-22-semaev-first-fall-replay`
  - The finite instance $n = n' = 4$ has operational first fall 3, where the literal Petit–Quisquater reading gives 2.
  - It contradicts no published theorem: Kosters–Yeo say "usually 2". It belongs as a remark in the main paper's definitions section, not as a paper of its own.

5. Considered and not recommended (index-calculus scope)

candidate	why not now
Plane-quartic (Diem) IC on GHS-weak $E_0/\mathbb{F}_{q^3}$ (`research/quartic_n4v2_20261020`)	Exponent ≈ 0.98 against rho's ≈ 1.5, but this is known: Diem 2003/2006 and Diem–Thomé 2008 ("faster than rho even for small group sizes"). Joux–Vitse 2012 solved 149-bit $\mathbb{F}_{p^6}$ curves. Also L0 host, B0 records do not replay, 3 units per size. Publishable only at record scale.
$k = 4$ D_4 Frobenius-reduced form, char 2 (`gaudry_quartic_d4_char2_20261015`)	The red team (`gaudry_k4_char2_redteam_20261026`) rated the crossover "NOT SURVIVED (unsupported)": 1–2 targets per rung. Invariant-ring work in char 2 is Galbraith–Gebregiyorgis 2014. The cheapest headline experiment (about 30 h, frozen plan) is worth running, but the paper waits on it.
Genus-3 hyperelliptic GHS with double large primes	Gaudry–Thomé–Thériault–Diem 2007 already beat rho in genus 3; the slope against rho is not robust (`ghs3_vs_rho_20261023`).
Root-finding decomposition (`papers/rootfinding-decomposition`)	Prop. 1 is the definition of S_3 (McGuire–Mueller; Amadori–Pintore–Sala). Engineering only.
Semaev monodromy / conservation (`crypto-autoresearcher:papers/semaev-conservation-specialization`)	The Galois group $(\mathbb{Z}/2)^{m-2}$ is plausibly folklore (signs of y_i); the paper itself says novelty is not adjudicated.

candidate	why not now
IC improvement ledger / programme record / audit methodology	Records, not venue papers; 0 of 28 IC leaderboard rows are validated.

6. Re-verification run for this report

To test gating item G1, I regenerated one certificate from the committed code and checked it with the independent checker.

- Instance T33a: $n = 33$, random $V \subset \ker \mathrm{Tr}$.
- Producer: `code/lfdclose2.c`, compiled against M4RI 20200125.
- Checker: `verify/cert_check.py`.
- Host and isolation: this cloud container (4 vCPU, 15 GB), L0. Only the verdict matters here, not the timing.

Result. The certificate reproduces **byte for byte**: SHA-256 `cf9dcae81bf1843fa55aeb5f5d3745b69a50790818b7eb48b16193` the value recorded in `results/certificates.txt`. The checker reports `VALID: 1 is not in W_4(F), and codim W_4(F) >= 4888`.

quantity	value
closure dimensions $\dim W_4 \cap B_{\leq d}$	(0, 0, 64, 2968, 48068), the recorded values
closure rounds	3
certificate size	33,044,576 bytes
closure time	4 min 40 s
check time	38.8 s

So gating item G1 is a matter of compute and hosting, not of lost code. The other certificates should regenerate the same way, at proportionally larger cost (P45a about 0.3 GB).

7. Method and limits

- **Inventories.** Four parallel read-only inventories: existing drafts; crypto research 2026-10-20…29; the crypto-autoresearcher lead atlas, Lean and ledger; and the cryptanalysis repository. A fifth, deeper read covered every fall-degree result.
- **Literature checks.** Done on the web; the ePrint PDF host refused direct download. Read in full: Kosters–Yeo (arXiv 1503.08001) and PKM 2016. Read from abstracts or secondary summaries only: the rest, including HKY 2015, Kousidis–Wiemers 2019 and FGHR. A submission needs G5.
- **Repository state.** The `cryptanalysis` checkout is at 2026-10-22: a fetch failed while the disk was full. Nothing in its fall-degree material is distinct from crypto's.
- **Not covered.** No reading of the 43 ECDLP ledger records that have no lead in the lead atlas.